What Does the Control Control For?

Estimand mismatch in activation patching, with evaluation awareness as the case study

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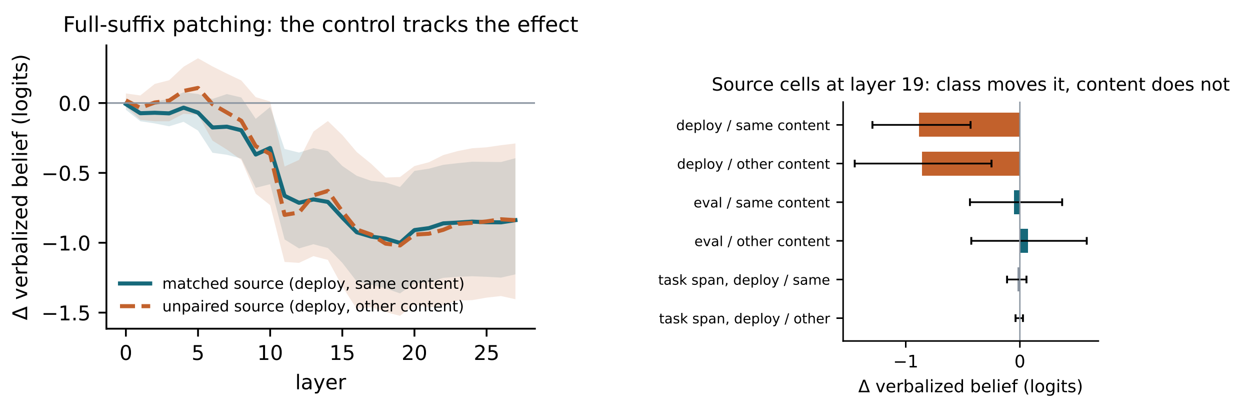
Executive summary

The question. Open-weight language models linearly encode whether they are being evaluated [5, 14]. Activation patching is the usual next step for turning that correlation into a causal claim. I asked whether patching demonstrates transfer of evaluation *context* — and whether the control I reached for measured what I thought it measured.

The observation that changed the project. Patching the shared suffix of a matched deployment prompt into an evaluation prompt moved the model's verbalized belief by -1.001 logits at deep layers. My control — the same patch sourced from an unrelated prompt — reproduced the entire curve (Fig. 1a). I initially read that as refutation. Both readings turn out to rest on an *estimand mismatch*: my paired and unpaired sources were both deployment-class prompts, so their difference held fixed exactly the factor I wanted to estimate, and could only ever measure content-matching effects.



(a) The observation (b) The diagnosis

Figure 1: (a) The unpaired control reproduces the matched patch effect (Qwen2.5-1.5B; 60 prompt pairs over 30 held-out content clusters; bands are content-clustered bootstrap CIs). (b) Completing the source design to a 2×2 of (class × content): both deploy-class sources (top two bars) produce the effect and both eval-class sources do not. My original contrast compared only the top two bars, and so cancelled the class effect.

The repaired analysis and the honest finding. With all four source cells, the estimated source-class effect at the readout span is -0.881 [-1.212, -0.531] logits; the content effect (0.072 [-0.346, 0.493]) and the class×content interaction (0.092 [-0.414, 0.620]) are not detectably different from zero. The class effect grows across Qwen2.5-{0.5B, 1.5B, 7B} while the paired-minus-unpaired contrast stays statistically indistinguishable from zero at every scale (Table 1). Splitting the patched region finer shows what the class signal is: on the primary model it is concentrated at the final pre-answer position (-0.82 [-1.10, -0.52]; reverse direction 0.87 [0.52, 1.20]; maximum-effect layer — exploratory, locked analog in Sec. 2), with the pre-question tokens contributing -0.04 [-0.08, -0.00]. Across five models the pattern is *predominantly* report construction, with smaller but detectable upstream transfer — an order of magnitude below the final-position effect — at the question and pre-question tokens on the larger Qwen models (Table 2). What patches move is mostly the state from which the model is about to read out its answer, not evaluation-context processing upstream

*Code, corpus, and every result file: [TO FILL: public repository URL]. Every number below regenerates from two commands. LLM assistance is disclosed in §8.

of it. Both my original readings — "causal localization of awareness" and "pure artifact" — were answers to a question my contrast never asked.

Strongest evidence against my interpretation. (i) The class effect could have reflected source–destination distance rather than class; a matched design now bounds this — with all four source cells at identical per-position residual distance and only shrunk offsets, the effect stands at -0.62 [-0.81, -0.44] (Sec. 2) — though the constructed estimand is the offset-direction effect at standardized dose, not the raw class effect re-measured. (ii) Upstream transfer is not absent: on Qwen-7B the question tokens carry $+0.24$ $[+0.11, +0.37]$ in one direction and the same-content pre-question contrast is detectable in both directions (-0.10 $[-0.18, -0.02]$ and $+0.16$ $[+0.06, +0.26]$), and on the primary model the question tokens carry -0.11 $[-0.18, -0.04]$, so "report construction" is a predominance claim, not an exclusivity claim. (iii) An all-layer direction-ablation experiment was unstable under its own random control (the control moved the outcome more than the treatment); I draw no conclusions from it. (iv) A second-concept replication (register) was null on the primary model and directional but not significant at 7B.

Narrowest honest limitation. One synthetic English corpus, 30 held-out content clusters, one readout question, models ≤8B; and the "class transfer" I localize is a property of a verbal report — I found no route from any of it to task behavior (sandbagging gaps null or marginal on all five models).

1 Setup, in brief

Corpus and models. 60 task items rendered under five evaluation and four deployment cue families (a no-lexicon exam family, structurally matched metadata families, topic controls), split 50/50 into train/test by content. Models: Qwen2.5-{0.5B, 1.5B, 7B}-Instruct, Phi-3-mini-4k-instruct, Olmo-3-7B-Instruct (TransformerLens, Apple silicon); primary model for in-text numbers: Qwen2.5-1.5B-Instruct in fp32. The representational result of prior work reproduces here (held-out probe AUC 0.956, permutation $p = 0.005$, computed at the selected layer and uncorrected for the max over layers (the permutation null's 95th percentile, 0.54 AUC, leaves any corrected version equally decisive); no-lexicon family transfer 0.995; topic-control false-positive rate 0.000 against 0.936 for genuine evaluation framings) and I claim no novelty for it [4, 5, 14].

Readout and direction. The verbalized belief is a two-polarity yes/no question appended after the task, scored from the first-answer-token logits at the final input position (positive = "I am being evaluated"); $d_\ell = \mu_\ell^{\text{eval}} - \mu_\ell^{\text{deploy}}$ (unit-normalized), steering doses in units of σ_ℓ. "Final pre-answer position" below means that last input position: no answer token has been generated when it is patched.

Data accounting. Layer sweeps: 60 pairs over 30 content clusters; all CIs are content-clustered bootstraps. Wherever a selected layer feeds a confirmatory estimate and that layer was among those run, I report a locked version — layer selected on a discovery half of the clusters, effect re-estimated with a clustered bootstrap on the other half ("locked"; 15 clusters each side; the one gap, Phi-3, is named in §7); layers labeled maximum-effect were selected on all data, and estimates at them are exploratory. *Attrition and its repair:* an earlier version of the 2×2 required different-content sources to share the readout span plus a two-token margin; on the Qwen and Olmo tokenizers those sources share the readout span plus exactly one glue token, so 34 of 60 candidate pairs were excluded (0 of 24 on Phi-3), leaving 13 clusters and model-dependent content subsets. The criterion was tokenizer-determined and effect-blind (`results/attrition.json`), but it made the models incomparable. All 2×2 estimates reported here therefore use one fixed list of the same 30 held-out contents on every model, with the margin removed (readout span fully shared is the only requirement). For different-content sources the content-specific pre-question tokens are not shared by construction (only glue tokens and, for multiple-choice items, the shared answer-format instruction survive the suffix match), so in the tabulated pre-question column the two different-content cells are near-empty patches and the 2×2 main effect understates the same-content pre-question contrast by roughly half (primary model: -0.07 $[-0.13, -0.01]$ same-content vs. -0.04 [-0.08, -0.00] tabulated).

2 The estimand mismatch

What my contrast estimated. Let $Y_{c,s}$ be the patch effect from a source of class $c \in \{\text{same}, \text{other}\}$ relative to the destination and content $s \in \{\text{same}, \text{other}\}$. The matched patch is $Y_{\text{other},\text{same}}$; my unpaired

model	estimated source-class effect (2x2 CI)	paired-minus-unpaired contrast (CI)
Qwen2.5-0.5B-Instruct	-0.39 [-0.47,-0.30]	-0.01 [-0.24,+0.21]
Qwen2.5-1.5B-Instruct	-0.88 [-1.21,-0.53]	-0.03 [-0.57,+0.50]
Qwen2.5-7B-Instruct	-2.46 [-3.38,-1.62]	+0.18 [-1.30,+1.87]
Phi-3-mini-4k-instruct	+0.00 [-0.00,+0.01]	-0.01 [-0.02,+0.01]
Olmo-3-7B-Instruct	-0.49 [-0.89,-0.12]	+0.01 [-0.81,+0.93]

Table 1: The paired-minus-unpaired contrast does not respond to the estimated source-class effect. Both columns: readout span at each model's maximum-patch-effect layer (selected on all data — exploratory; the locked estimates are in Table 4), the same 30 content clusters on every model, content-clustered bootstrap CIs.

control was $Y_{\text{other,other}}$. Their difference holds source class fixed: it estimates the content-matching effect plus the class×content interaction and is structurally incapable of estimating a class effect. I make no claim that the literature formally defines this contrast as a class estimator; what I am reporting is the inference I made, which I believe is tempting for anyone who reaches for an unrelated-prompt control [6, 19] — read a large matched effect as concept transfer, read a matching control as refutation — and which the design cannot license either way. It is distinct from corruption-choice sensitivity [19] and from subspace-patching illusions [10]; the nearest analogue is over-control.

The full factorial. Adding the two same-class cells gives, at the primary model's maximum-effect layer and readout span: estimated source-class effect -0.881 [-1.212, -0.531]; content effect 0.072 [-0.346, 0.493]; interaction 0.092 [-0.414, 0.620] (not detectable — the condition under which the simpler paired-minus-same-class contrast, -0.835 [-1.288, -0.366], is a clean class estimate). Because that layer was chosen on all data, I also report a locked estimate: the patch layer selected on the discovery half of the contents (layer 19, which coincides with the all-data choice), estimated on the estimation-half contents only: -1.028 [-1.550, -0.457] (15 clusters). Table 1 repeats the decomposition on five models: the class effect spans a wide range while the paired-minus-unpaired contrast never leaves a zero-straddling band. The class effect is depth-ordered ($\rho = 0.97$) and concentrates late — the mismatch bites hardest exactly where naive patching looks most impressive.

Where the class signal sits. Splitting the patched region — pre-question task tokens, belief-question tokens, final pre-answer position — in both patch directions on all five models (Table 2) gives a predominance result, not an exclusivity result. On the primary model the signal is concentrated at the final pre-answer position (-0.82 [-1.10, -0.52] vs. question -0.11 [-0.18, -0.04] and pre-question -0.04 [-0.08, -0.00]); the locked analog (discovery-half layer, estimation-half clusters) gives -0.920 [-1.372, -0.404] and, in the reverse direction, 0.729 [0.216, 1.200]; the locus differs by model (question tokens on the smallest Qwen and on Olmo; final pre-answer position on the larger Qwen models; every cell zero on Phi-3), and smaller upstream transfer is detectable on the larger Qwen models — always an order of magnitude below the dominant position. The interpretation I consider honest: patching mostly transplants the state from which the model is about to read out its report — constructed by the final pre-answer position on larger Qwen models, still distributed over the question tokens at 0.5B — with a minority of transfer that is genuinely upstream. Olmo also splits the region in a way that explains its steering anomaly (§4): large normal-signed transfer at the question tokens and, in one patch direction, sign-inverted transfer at the final pre-answer position, so instruments that weight positions differently must disagree.

Distance is matched by construction. Per-cell source–destination residual distances were recorded for every patch, and observationally they cut against a distance account (the inert same-class/different-content source lies *farther*, $\ell_2 = 12.8$, than the potent other-class/same-content source, 9.2). Because that is a check and not a design, I re-ran the full 2×2 with distance handled by construction (Experiment J): same destinations, patched spans equalized across the four cells, same clustered estimator, 30 clusters per destination class (unmatched baseline under this discipline: -0.82 [-1.10, -0.52] / 0.87 [0.52, 1.20], reproducing the published estimates). Distances are functions of activations fixed before any intervention, so matching on them is outcome-blind covariate matching; exact balance among *real* activations is nevertheless geometrically impossible if class is linearly encoded (an other-class source has a distance floor), so selection serves as the on-manifold check and construction as the primary instrument. *Distance-selected* (best of 6 candidate

model	destination	pre-question	question	final pre-answer
Qwen2.5-0.5B-Instruct	eval dest.	-0.00 [-0.03,+0.03]	-0.41 [-0.50,-0.33]	+0.02 [-0.02,+0.06]
Qwen2.5-0.5B-Instruct	deploy dest.	+0.01 [-0.02,+0.04]	+0.44 [+0.32,+0.56]	-0.01 [-0.04,+0.02]
Qwen2.5-1.5B-Instruct	eval dest.	-0.04 [-0.08,-0.00]	-0.11 [-0.18,-0.04]	-0.82 [-1.10,-0.52]
Qwen2.5-1.5B-Instruct	deploy dest.	+0.02 [-0.03,+0.07]	+0.06 [-0.03,+0.14]	+0.87 [+0.52,+1.20]
Qwen2.5-7B-Instruct	eval dest.	-0.05 [-0.10,+0.01]	-0.19 [-0.31,-0.06]	-2.27 [-3.12,-1.47]
Qwen2.5-7B-Instruct	deploy dest.	+0.12 [+0.06,+0.19]	+0.24 [+0.11,+0.37]	+3.24 [+2.33,+4.18]
Phi-3-mini-4k-instruct	eval dest.	+0.00 [-0.01,+0.01]	-0.00 [-0.00,+0.00]	+0.00 [-0.00,+0.00]
Phi-3-mini-4k-instruct	deploy dest.	-0.02 [-0.04,+0.00]	-0.00 [-0.01,+0.00]	-0.00 [-0.01,+0.00]
Olmo-3-7B-Instruct	eval dest.	-0.00 [-0.09,+0.08]	-0.93 [-1.15,-0.71]	+0.10 [-0.17,+0.37]
Olmo-3-7B-Instruct	deploy dest.	+0.11 [-0.03,+0.28]	+0.75 [+0.54,+0.94]	-0.50 [-0.91,-0.09]

Table 2: Estimated source-class effect by patched position, both patch directions, all five models (maximum-effect layer, selected on all data — exploratory; locked final-position analogs in the text; the same 30 content clusters per row on every model; content-clustered CIs).

renderings per cell, chosen — before any effect is computed — to zero the class-condition distance gap; achieved gap -0.00 vs. -0.24 unmatched, eval destinations): class effect -0.95 [-1.14, -0.77] / 0.93 [0.65, 1.21]. *Distance-constructed* (each source's per-position offset rescaled to the four-cell mean norm, so all four cells sit at identical ℓ_2 distance at every patched position; mean rescale 1.14): -0.87 [-1.16, -0.57] / 0.87 [0.56, 1.16]. *Shrink-only construction* (rescale to the four-cell *minimum* norm: every patched state is a convex interpolation toward the destination, no offset is ever amplified, so rescaling can only attenuate whatever the offset carries — the conservative primary, decision rule committed before the runs): -0.62 [-0.81, -0.44] / 0.61 [0.40, 0.81]. The locked versions (discovery-half layer 19, estimation-half clusters, $n = 15$) stand: shrink-only -0.690 [-0.976, -0.386], selected -1.031 [-1.278, -0.792]. Balance also holds in a unit-direction metric (matched-arm class-condition gaps below 0.005 in every cell on every model and ≤ 0.003 at the final position; 0.002 and 0.001 on the primary model), so RMSNorm radial invariance does not hide an imbalance, and the rescale footprint — the paired unmatched-minus-constructed change in each cell's own effect — stays within 0.59 logits across all cells, spans and layers on the primary model (0.53 at the final position) and within 1.85 logits on any of the five, bounding what off-manifold distortion could contribute. Two scope notes: the constructed arms estimate the effect of the class-associated offset *direction* at standardized per-position dose (equalizing magnitude removes any magnitude-mediated share of the raw effect along with the confounded share), and the selected arm re-matches per analysis cell; all arms share one survivor set of destinations, and the constructed arms reuse the unmatched sources. At matched — and at identical — residual distance, class still orders the effect (Table 3). On the two models whose final-position class effect is detectable in *both* patch directions the shrink-only estimate attenuates relative to the unmatched one (1.5B $-0.82 \to -0.62$; 7B $-2.27 \to -1.26$) while excluding zero in both patch directions: part of the raw effect does travel with offset magnitude, and the majority does not. The result is not confined to that position. Qwen2.5-0.5B carries no final-position effect to match — its locus is the belief-question span (§2, Table 2) — and matched over the readout span that contains that locus it replicates too: shrink-only -0.25 $[-0.32, -0.18]$ and, in the reverse direction, $+0.15$ $[+0.09, +0.22]$ (Experiment J resolves final, readout and pre-question spans, not the question tokens alone). Table 3 therefore carries a readout-span column, so a model whose class effect does not sit at the final position is not represented by a row of zeros.

Where matching removes an effect. Olmo-3-7B is the case the matched design argues against itself, and it is worth stating plainly. Its anomalous *sign-inverted* final-position effect — the one that mechanically explains its inverted steering (§4) — does not survive the conservative arm: unmatched -0.50 $[-0.91, -0.09]$ (deploy destinations) becomes -0.34 $[-0.59, -0.09]$ when matched by selection, -0.38 $[-0.75, -0.00]$ when constructed, and -0.07 $[-0.25, +0.11]$ under shrink-only construction, which no longer excludes zero. I read that inversion as partly a magnitude phenomenon rather than class transfer, and I do not claim it as a matched result. What does survive on Olmo is the effect at its own locus, the belief-question tokens, and it survives *normal*-signed in both directions (shrink-only -0.39 $[-0.61, -0.15]$ and $+0.53$ $[+0.27, +0.81]$ over the readout span). Phi-3-mini, which has no detectable class effect anywhere in the unmatched design, has none under matching either: every arm, span and direction stays within 0.01 logits of zero at the maximum-effect layer and 0.04 at every layer run — a calibration check on the instrument rather than a replication. The pre-question task span stays *small* under matching on all five models — every matched estimate at the

model	destination	unmatched	distance-selected	distance-constructed	shrink-constructed	shrink, re
Qwen2.5-0.5B-Instruct	eval dest.	+0.02 [-0.02,+0.06]	+0.02 [-0.02,+0.07]	+0.02 [-0.03,+0.07]	+0.02 [-0.01,+0.04]	-0.25 [-0.3
Qwen2.5-0.5B-Instruct	deploy dest.	-0.01 [-0.04,+0.02]	-0.06 [-0.10,-0.02]	-0.01 [-0.06,+0.03]	-0.01 [-0.03,+0.01]	+0.15 [+0
Qwen2.5-1.5B-Instruct	eval dest.	-0.82 [-1.10,-0.52]	-0.95 [-1.14,-0.77]	-0.87 [-1.16,-0.57]	-0.62 [-0.81,-0.44]	-0.69 [-0.9
Qwen2.5-1.5B-Instruct	deploy dest.	+0.87 [+0.52,+1.20]	+0.93 [+0.65,+1.21]	+0.87 [+0.56,+1.16]	+0.61 [+0.40,+0.81]	+0.67 [+0
Qwen2.5-7B-Instruct	eval dest.	-2.27 [-3.12,-1.47]	-1.81 [-2.35,-1.17]	-1.94 [-2.70,-1.15]	-1.26 [-1.75,-0.75]	-1.37 [-1.9
Qwen2.5-7B-Instruct	deploy dest.	+3.24 [+2.33,+4.18]	+2.11 [+1.32,+2.98]	+2.82 [+1.94,+3.72]	+1.46 [+1.02,+1.94]	+1.52 [+1
Phi-3-mini-4k-instruct	eval dest.	+0.00 [-0.00,+0.00]	+0.00 [-0.00,+0.01]	+0.00 [-0.00,+0.00]	+0.00 [-0.00,+0.00]	-0.00 [-0.0
Phi-3-mini-4k-instruct	deploy dest.	-0.00 [-0.01,+0.00]	-0.00 [-0.01,-0.00]	-0.00 [-0.00,+0.00]	-0.00 [-0.00,+0.00]	-0.00 [-0.0
Olmo-3-7B-Instruct	eval dest.	+0.10 [-0.17,+0.37]	-0.03 [-0.20,+0.14]	+0.07 [-0.18,+0.33]	+0.09 [-0.08,+0.28]	-0.39 [-0.6
Olmo-3-7B-Instruct	deploy dest.	-0.50 [-0.91,-0.09]	-0.34 [-0.59,-0.09]	-0.38 [-0.75,-0.00]	-0.07 [-0.25,+0.11]	+0.53 [+0

Table 3: Distance-matched estimated source-class effects, both patch directions, for every model with a completed Experiment J run. The four estimate columns after the destination are the final pre-answer position; the last is the shrink-only estimate over the whole readout span (belief question + final position), which contains every model's locus and so remains informative for models whose class effect does not sit at the final position. Maximum-effect layer — exploratory; the locked shrink-only estimate is in the text; content-clustered CIs.

model	probe AUC	verbal AUC	sandbag gap (paired CI)	locked steering (eval framing)	class effect, max-effect layer (exploratory)	class effect, locked (disc.-half layer, est.-half contents)
Qwen2.5-0.5B-Instruct	0.967	0.835	+0.03 [-0.06,+0.14]	L14: -0.29 [-0.32,-0.25]	-0.39 [-0.47,-0.30] (n=30)	-0.35 [-0.45,-0.25] (n=15)
Qwen2.5-1.5B-Instruct	0.956	0.783	+0.02 [-0.10,+0.15]	L10: -0.84 [-1.03,-0.67]	-0.88 [-1.21,-0.53] (n=30)	-1.03 [-1.55,-0.46] (n=15)
Qwen2.5-7B-Instruct	0.977	0.885	-0.00 [-0.03,+0.03]	L13: -2.20 [-2.53,-1.87]	-2.46 [-3.38,-1.62] (n=30)	-2.72 [-4.31,-1.47] (n=15)
Phi-3-mini-4k-instruct	0.981	0.698	+0.01 [-0.03,+0.08]	L12: -0.01 [-0.05,+0.03]	+0.00 [-0.00,+0.01] (n=30)	–
Olmo-3-7B-Instruct	0.994	0.643	+0.05 [-0.00,+0.13]	L14: +0.32 [+0.12,+0.53]	-0.49 [-0.89,-0.12] (n=30)	-0.52 [-1.11,-0.04] (n=15)

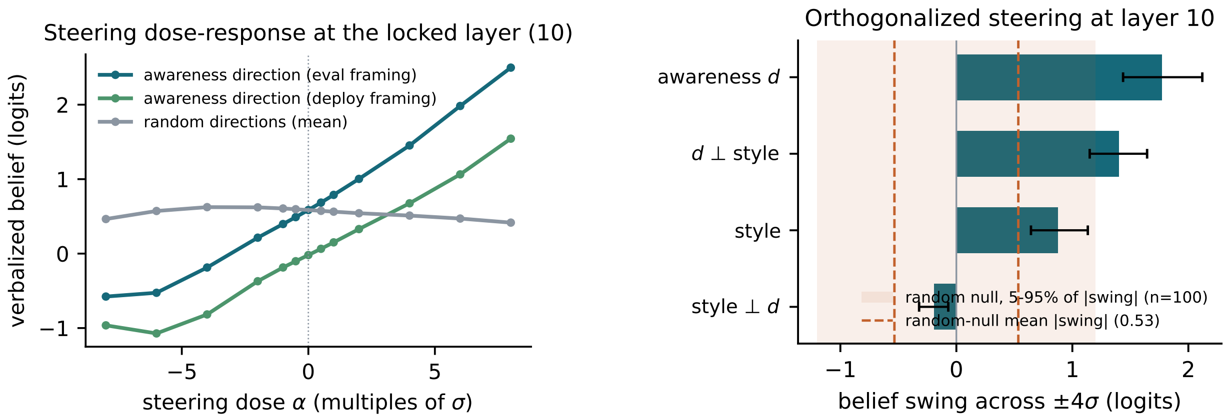
Table 4: Five models, identical instruments and identical content clusters. "Locked steering": held-out eval-framing estimate at the discovery-selected layer. Class effects: the 2×2 estimate at the readout span, at the all-data maximum-effect layer (exploratory) and locked (layer selected on the discovery half of contents, estimated on the other half). Sandbagging CIs are paired and content-clustered (n = 24 items per model).

maximum-effect layer lies within 0.05 logits of zero (largest, Qwen-7B shrink-only deploy destinations, -0.05 $[-0.09, -0.01]$) — but several of those small values do exclude zero, so this reproduces the predominance claim of §2 under matching, not an exclusivity claim. All five checkpoints have now completed Experiment J on the same 30 content clusters per destination class.

Scope. The same 2×2 on a second concept (register, formality readout) gave a null class effect at 1.5B (-0.08 $[-0.45, +0.25]$) and a directional but non-significant one at 7B (-0.41 $[-1.09, +0.26]$; 12 contents), so the estimand point generalizes by its logic — a contrast that fixes the concept label cannot estimate that label's effect — rather than by a second demonstrated instance.

3 Convergent evidence from steering

Steering $\pm\alpha\sigma_\ell\hat{d}_\ell$ transplants no span. Held to a pre-specified battery — a locked layer (selected on the discovery split; on the primary model it coincides with the all-data steering peak, at which the dose and percentile diagnostics below are computed), dense dose-response, distributional checks (KL from the unsteered next-token distribution at every dose), and a null of 100 matched-norm random directions over 30 clusters — steering on the primary model moves the verbalized belief at the locked layer by -0.844 [-1.026, -0.673] (eval framing) and 0.783 [0.630, 0.932] (deploy framing). I treat the low-KL regime ($|\alpha| \leq 1$, KL ≤ 0.009 nats) as primary: there the swing is 0.439 [0.346, 0.539] and $|d|$ sits at the 98th percentile of the random-direction null (Fig. 2a). Large doses (mean KL 0.68 nats at $|\alpha| = 8$, averaged over framings) are reported as secondary. Steering and the repaired patching estimator agree in sign and rough magnitude on the Qwen family — convergent evidence that the class variable is real and manipulable — while remaining, like patching, interventions on what the model *reports*: none of them measurably changed task behavior (§4).



(a) Dose-response at the locked layer

(b) Orthogonalized steering vs. the random null

Figure 2: (a) Steering dose-response at the locked layer; random-direction mean for comparison. (b) Matched-norm swings for the awareness direction, the style direction, and each orthogonalized to the other, against the 5–95% band of |swing| over 100 random directions (secondary result, §5).

4 Instrument validity across five models

The encoding side is consistent everywhere (probe AUC 0.956–0.994 with format controls passing; the readout region dominates causal span maps in all five). The causal side is not. On Qwen, locked steering grows with scale ($-0.29 \rightarrow -0.84 \rightarrow -2.20$) and the source-class effect grows with it. On Phi-3-mini nothing meaningfully moves the report: the largest eval-framing steering effect at any layer is -0.030 $[-0.044, -0.016]$ (deploy framing $+0.036$ $[+0.025, +0.049]$) — two orders of magnitude below Qwen-7B — and every $2{\times}2$ cell is zero. On Olmo-3-7B, locked steering CIs exclude zero ($+0.32$ $[+0.12, +0.53]$ eval framing) with *inverted* signs, and orthogonalized steering shows d, style, and random directions all moving the readout comparably — the inverted-steering regime of Torop et al. [16], arising in the wild; effect-size-with-CI reporting would have passed it as replication. The positional split of §2 explains the configuration (question tokens normal, final pre-answer position inverted in one direction), and the $2{\times}2$ still detects class transfer on Olmo. Instruments dissociate on the same model; that is the strongest argument I know for running all of them.

Behavior, for context and consistent with prior work [8, 9]: incentivized sandbagging gaps are null with bounded CIs on four models (Qwen-7B: -0.00 $[-0.03, +0.03]$; pooled Qwen: 0.003 [-0.026, 0.032]) and marginal on Olmo ($+0.054$ $[-0.002, +0.126]$); steering the belief direction hard enough to move the report by a logit moves accuracy by 0.001.

5 Secondary result: the register share of the steerable signal

A style direction — the mass-mean of formal-vs-casual framings, built from deployment-side prompts only — is substantially aligned with d in every model (cos = 0.46–0.68 at the steering layer). Steering with d, style, and each orthogonalized to the other at matched norm (Fig. 2b) shows the style axis going from causally inert at 0.5B (2% of d's swing; $d_{\perp\text{style}}$ keeps 97%) to parity with d at 7B (64% of d surviving orthogonalization; in the low-KL regime on the primary model, 78%). I report this as a within-family trend with caveats, not a scaling law: three sizes of one family; a style direction constructed from my own deployment framings, which is *correlated with*, not guaranteed free of, evaluation-context information; and a random-direction floor that is a substantial fraction of d's swing at small scale. Its value is as a causal-level counterpart to the probe-level format-sensitivity critique [4]: belief-report controllability claims need register controls.

6 Related work

Evaluation awareness: decodability, steering, scaling, and dissociations [1, 3, 5, 7–9, 11–14]; format-sensitivity of probes [4]. Patching methodology: corruption-type sensitivity [19], practice [6, 18], subspace illusions [10]; the estimand of the control contrast is orthogonal to all three. Steering validity: inverted steering vectors [16]; steering practice [2, 15, 17, 20].

7 Limitations, and evidence against

Report-construction predominance is a predominance claim: upstream transfer is detectable at the question and pre-question tokens on the larger Qwen models, an order of magnitude below the dominant position. Content and interaction "nulls" are bounded, not negligible: a TOST-style equivalence analysis (90% clustered bootstrap CIs; final position, maximum-effect layer; `scripts/equivalence_bounds.py`) bounds the primary model's content and interaction effects within 0.42–0.63 logits, i.e. 0.51–0.72× the class effect (both destinations), with 0.47–0.75× on Qwen-7B deploy destinations; on Olmo, and wherever the final-position class effect is itself undetectable, no informative equivalence claim is supportable. One cell is not a null at all: the Qwen-7B eval-destination interaction at the final position is detectably positive (+1.93 [+0.35, +3.68], 95%), so the same-content class contrast there is weaker than the different-content one. The maximum-effect layer was selected on all data; the locked estimates (discovery-half layer, estimation-half contents) agree but use 15 clusters, and are not computable for Phi-3 (whose discovery-selected layer was not among those run). An all-layer direction-ablation experiment was unstable under its own random control (control -0.780 [-1.116, -0.481], treatment 0.085 [-0.255, 0.435], not distinguishable from zero) and is excluded from all claims. The register replication did not reach significance. Behavioral nulls are bounded by 24-item suites. One corpus, synthetic, English; five models $\leq$8B from three labs.

8 Disclosure and verification

Tools. Code, experiment orchestration, and drafting were done with substantial assistance from Anthropic's Claude (Claude Code); the framing of the estimand critique and the list of blocking issues addressed in this version came from written critiques produced with OpenAI's ChatGPT/Codex. Multi-agent LLM audits cross-checked prose against result files; the errors they found are corrected here, and I do not treat them as a substitute for the personal verification below.

What I personally designed, read, and checked. [TO FILL: which hypotheses and controls you designed yourself (e.g. the 2×2 completion, the locked split, the positional split)]. [TO FILL: which code paths you read end to end (e.g. patching hooks in `interventions.py` and `exp7/exp10`; the belief readout in `readouts.py`)]. [TO FILL: how many raw prompts and readouts you inspected by hand (`scripts/sample_audit.py`), and whether alignments and scores looked right]. [TO FILL: which headline numbers you independently recomputed from the CSVs (e.g. the class main effect, the interaction, one locked steering CI)]. [TO FILL: what remains unchecked]. [TO FILL: for each component — corpus, patching, steering, statistics — how surprised you would be by a major error, in one line each]. [TO FILL: total active hours, and a short decision timeline]. [TO FILL: exact contribution of anyone else who participated, or "none"].

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